

Course Code: CCE-332

Course Title: Peripheral and Interfacing Sessional

Group: 07

Comparison with related works-

Name	Primary Function(s)	Locomotion & Control	Key Features & Innovations
EMO Robot (Our Project)	Real-time facial emotion detection and interactive robot response	Fixed robot body, Raspberry Pi 3, Camera, Servo Motor, Python controller.	Detects emotions (happy, sad, angry, neutral), shows output via OLED display, LED color, servo movement, and speaker feedback in real time.
Multimodal Emotion Detection System [1]	Detects human emotion using multiple modalities	Computer-based HRI system	Uses facial expressions, speech, and physiological signals for emotion recognition; improves human-robot interaction.
Emotion Recognition for Human Robot Interaction [2]	Emotion classification for social robots	AI-based robot control	Reviews facial, voice, body-pose based recognition methods; focuses on improving robot social intelligence.
Robotics and Emotion Survey [3]	Emotion-aware robotic behavior	Social robot framework	Large-scale survey of emotion-based robot systems and interaction models.
Multimodal Fusion for Human Robot Interaction [4]	Emotion recognition using deep learning	Intelligent robot system	Combines multiple sensor inputs and multimodal fusion for better recognition accuracy.
Bangla Speech Emotion Recognition [5]	Speech-based emotion detection	Deep learning model	Deep learning model.

References

- [1] L. Zhang, M. Jiang, D. Farid, and A. Hossain, "Intelligent facial emotion recognition and semantic-based topic detection for a humanoid robot," *Expert Systems with Applications*, vol. 40, no. 13, pp. 5160–5168, 2013.
- [2] M. Spezialetti, G. Placidi, and S. Rossi, "Emotion recognition for human-robot interaction: Recent advances and future perspectives," *Frontiers in Robotics and AI*, vol. 7, Art. no. 532279, 2020.
- [3] A. Author *et al.*, "Multi-face Emotion Detection for Effective Human-Robot Interaction," *arXiv preprint*, arXiv:2501.07213, Jan. 2025.
- [4] [3] R. Raj, "An Improved Facial Emotion Recognition System Using Deep Learning," *Scientific Reports*, vol. 15, 2025.
- [5] M. T. A. Tashrif, J. M. A. Khan, S. H. Mahir, and M. S. Uddin, "Emotion recognition from Bangla dialect speech using privacy-aware deep learning models: A comparative analysis," *International Journal of Speech Technology*, vol. 29, pp. 41–58, 2026.